NumPy vs Pandas — Working with Data

Topics: NumPy arrays, Pandas DataFrames, basic operations, comparison of use cases

Goal

Learn the difference between **NumPy** and **Pandas**, how each one represents data, and when to use each library.

Context

In data analysis and Machine Learning, raw data often starts as numbers, then grows into structured tables with columns and labels.

NumPy focuses on **fast numerical computation**, while Pandas focuses on **labeled and structured data**.

Step 1: Working with NumPy

NumPy is mainly used for numerical arrays and mathematical operations.

```
import numpy as np

scores = np.array([70, 85, 90, 60, 75])

print(scores)
print("Average:", scores.mean())
print("Max:", scores.max())
```

NumPy arrays:

- Contain values of the same type
- Are very fast for math operations
- Do not have column names

Step 2: Working with Pandas

Pandas is built on top of NumPy and is used for tabular data.

```
import pandas as pd

data = {
    "name": ["Ali", "Sara", "Omar", "Lina"],
    "score": [70, 85, 90, 60]
}

df = pd.DataFrame(data)

print(df)
print("Average score:", df["score"].mean())
```

Pandas DataFrames:

- Have column names
- Can mix data types
- Are easy to filter and inspect

Step 3: Loading Data from a File

Pandas is especially useful when working with files.

```
import pandas as pd

df = pd.read_csv("dataset.csv")

print(df.head())
print(df.describe())
```

This is a common step before training Machine Learning models.

Step 4: Same Task — Two Approaches

Compare how NumPy and Pandas handle the same task: computing an average.

NumPy

```
avg = scores.mean()
```

Pandas

```
avg = df["score"].mean()
```

When to Use Each One

Use Case	NumPy	Pandas
Pure numerical computation	✔	✖
Labeled columns	✖	✔
Reading CSV files	✖	✔
Fast math operations	✔	✔ (via NumPy)

Your Task

- Load `dataset.csv` using Pandas
- Extract one numeric column as a NumPy array
- Compute the average using:
  - NumPy
  - Pandas
- Print both results

Expected Outcome

You should understand:

- What NumPy is good at
- What Pandas is good at
- Why Pandas is usually used before ML models
- Why NumPy is used inside ML algorithms

Focus on understanding the tools, not memorizing syntax.